

8 **(a)** The matrix **M** is $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$.

(i) Find $\mathbf{M}^2$. **[1]**

(ii) Write down the transformation represented by **M**. **[1]**

(iii) Hence state the geometrical significance of the result of part **(i)**. **[1]**

(b) The matrix **N** is $\begin{pmatrix} k+1 & 0 \\ k & k+2 \end{pmatrix}$, where k is a constant.

Using determinants, investigate whether **N** can represent a reflection. **[4]**